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MODELLING AND CONTROL OF WHEEL SHIMMY

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Introduction

This report will present the results for all the tasks included in the Modelling and Control of Multibody Mechanical Systems 2016 coursework, Modelling and Control of Wheel Shimmy. The report will end with a conclusion on the whole coursework and important points that can be learnt from it. The Matlab commands used for some of the tasks are shown in the appendix.

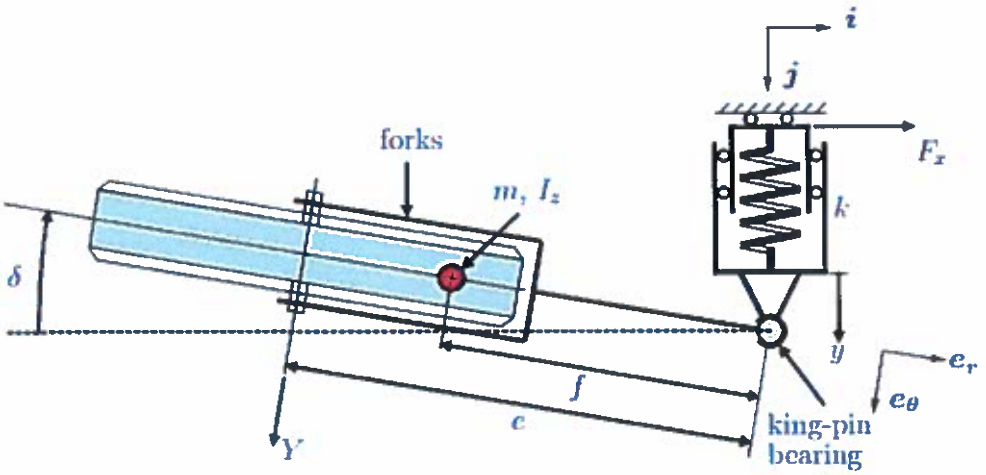


Figure 1: Plan (top) view of a simple system capable of shimmy

Question 1

Assume that the tyre can sideslip and the tyre force is lagged. Derive the equations of motion by hand:

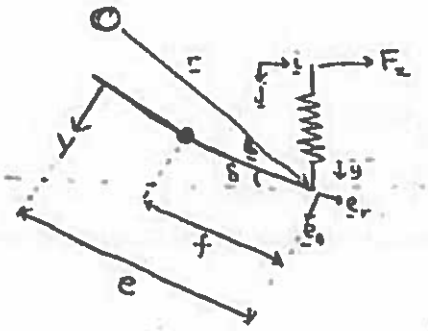


Figure 2: A simpler drawing of the system in Figure 1 with vectors used in the derivation

$\boldsymbol{R}$ *is the position vector of the centre of mass*

$$\boldsymbol{R} = x\boldsymbol{i} + y\boldsymbol{j} - f\boldsymbol{e_r}$$

$$\dot{\boldsymbol{R}} = \dot{x}\boldsymbol{i} + \dot{y}\boldsymbol{j} - f\dot{\delta}\boldsymbol{e_\theta}$$

$$\boldsymbol{e_\theta} = -\sin\delta\,\boldsymbol{i} + \cos\delta\,\boldsymbol{j} \qquad \text{and} \qquad \boldsymbol{e_r} = \cos\delta\,\boldsymbol{i} + \sin\delta\,\boldsymbol{j}$$

Therefore:

$$\dot{\boldsymbol{R}} = \dot{x}\boldsymbol{i} + \dot{y}\boldsymbol{j} - f\dot{\delta}(-\sin\delta\,\boldsymbol{i} + \cos\delta\,\boldsymbol{j})$$

$$\dot{\boldsymbol{R}} = (\dot{x} + f\dot{\delta}\sin\delta)\boldsymbol{i} + (\dot{y} - f\dot{\delta}\cos\delta)\boldsymbol{j}$$

The kinetic energy of the system:

$$T = \frac{1}{2}m\dot{\boldsymbol{R}}\cdot\dot{\boldsymbol{R}} + \frac{1}{2}I_z\dot{\delta}^2$$

Substitute $\dot{\boldsymbol{R}}$:

$$T = \frac{1}{2}m\left[(\dot{x} + f\dot{\delta}\sin\delta)^2 + (\dot{y} - f\dot{\delta}\cos\delta)^2\right] + \frac{1}{2}I_z\dot{\delta}^2$$

$$T = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + 2\dot{x}f\dot{\delta}\sin\delta - 2\dot{y}f\dot{\delta}\cos\delta + f^2\dot{\delta}^2) + \frac{1}{2}I_z\dot{\delta}^2$$

The potential energy of the system:

$$V = \frac{1}{2}ky^2$$

Using the Lagrangian approach:

$$L = T - V = \frac{1}{2}m\left(\dot{x}^2 + \dot{y}^2 + 2\dot{x}f\dot{\delta}\sin\delta - 2\dot{y}f\dot{\delta}\cos\delta + f^2\dot{\delta}^2\right) + \frac{1}{2}I_z\dot{\delta}^2 - \frac{1}{2}ky^2$$

The Lagrangian equation with respect to the generalised coordinate x is:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) - \frac{\partial L}{\partial x} = F_x - Y\sin\delta$$

$$\frac{d}{dt}\left(m(\dot{x} + f\dot{\delta}\sin\delta)\right) - 0 = F_x - Y\sin\delta$$

Therefore, the first equation of motion is:

$$m\left(\ddot{x} + f\ddot{\delta}\sin\delta + f\dot{\delta}^2\cos\delta\right) = F_x - Y\sin\delta \qquad (1)$$

The Lagrangian equation with respect to the generalised coordinate y is:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{y}}\right) - \frac{\partial L}{\partial y} = Y\cos\delta$$

$$\frac{d}{dt}\left(m(\dot{y} - f\dot{\delta}\cos\delta)\right) + ky = Y\cos\delta$$

Therefore, the second equation of motion is as follows:

$$m\left(\ddot{y} - f\ddot{\delta}\cos\delta + f\dot{\delta}^2\sin\delta\right) = -ky + Y\cos\delta \qquad (2)$$

And finally, the Lagrangian equation with respect to the generalised coordinate δ is:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\delta}}\right) - \frac{\partial L}{\partial \delta} = -eY$$

$$\frac{d}{dt}\left[m(\dot{x}f\sin\delta - \dot{y}f\cos\delta + f^2\dot{\delta}) + I_z\dot{\delta}\right] - m\left(\dot{x}f\dot{\delta}\cos\delta + \dot{y}f\dot{\delta}\sin\delta\right) = -eY$$

Therefore, the third equation of motion is:

$$m\left(f\ddot{x}\sin\delta - f\ddot{y}\cos\delta + f^2\ddot{\delta}\right) + I_z\ddot{\delta} = -eY \qquad (3)$$

The Newtonian approach to obtain the same equations are shown in Appendix A for comparison.

Question 2

The forward speed of the king-pin, $\dot{x}$, is a constant value, v by applying a force F_x. What should F_x be?

From (1):

$$F_x = m\left(\ddot{x} + f\ddot{\delta}\sin\delta + f\dot{\delta}^2\cos\delta\right) + Y\sin\delta$$

If $\dot{x}$ is constant, then $\ddot{x} = 0$

Therefore:

$$F_x = m\left(f\ddot{\delta}\sin\delta + f\dot{\delta}^2\cos\delta\right) + Y\sin\delta$$

Summary of Comments on Modelling and control of wheel shimmy

Page: 4

Number: 1 Author: Simos Evangelou Date: Indeterminate
Simos Evangelou: What about the equation with respect to the tyre lateral force, Y?

Question 3

Assume that, in addition to $\dot{x}$ being constrained, the tyre is also subjected to a nonholonomic constraint. Derive the new equations of motion:

The velocity vector at contact point:

$$\dot{r}_c = \dot{x}\boldsymbol{i} + \dot{y}\boldsymbol{j} - e\dot{\delta}\boldsymbol{e_\theta}$$

$$= \dot{x}(-\sin\delta\, \boldsymbol{e_\theta} + \cos\delta\, \boldsymbol{e_r}) + \dot{y}(\cos\delta\, \boldsymbol{e_\theta} + \sin\delta\, \boldsymbol{e_r}) - e\dot{\delta}\boldsymbol{e_\theta}$$

Due to the nonholonomic constraint, $v_{lat} = 0$

$$v_{lat} = -\dot{x}\sin\delta + \dot{y}\cos\delta - e\dot{\delta} = 0$$

Therefore, the first equation of motion is:

$$-v\sin\delta + \dot{y}\cos\delta - e\dot{\delta} = 0 \qquad (1)$$

Since $\dot{x} = v$, the Lagrangian can be simplified to:

$$L = T - V = \frac{1}{2}m\left(v^2 + \dot{y}^2 + 2vf\dot{\delta}\sin\delta - 2\dot{y}f\dot{\delta}\cos\delta + f^2\dot{\delta}^2\right) + \frac{1}{2}I_z\dot{\delta}^2 - \frac{1}{2}ky^2$$

Using the Lagrangian λ-method, the Lagrangian equation with respect to y is as follows:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{y}}\right) - \frac{\partial L}{\partial y} + \lambda\cos\delta = 0$$

$$\frac{d}{dt}\left(m(\dot{y} - f\dot{\delta}\cos\delta)\right) + ky + \lambda\cos\delta = 0$$

Therefore, the second equation of motion is as follows:

$$m\left(\ddot{y} - f\ddot{\delta}\cos\delta + f\dot{\delta}^2\sin\delta\right) = -ky - \lambda\cos\delta \qquad (2)$$

The Lagrangian equation with respect to δ is now:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\delta}}\right) - \frac{\partial L}{\partial \delta} - e\lambda = 0$$

$$\frac{d}{dt}\left[m(vf\sin\delta - \dot{y}f\cos\delta + f^2\dot{\delta}) + I_z\dot{\delta}\right] - m\left(vf\dot{\delta}\cos\delta + \dot{y}f\dot{\delta}\sin\delta\right) - e\lambda = 0$$

Therefore, the third equation of motion is:

$$m\left(-f\ddot{y}\cos\delta + f^2\ddot{\delta}\right) + I_z\ddot{\delta} = e\lambda \qquad (3)$$

1

Page: 5

Number: 1 Author: Simos Evangelou Date: Indeterminate

Simos Evangelou: You have essentially solved the problem. Eqs. 2 and 3 could have been used to find an equation for lambda (which gives the expression for the force of constraint) and an equation independent of lambda, which is the equation that describes the motion of the system together with the constraint equation (Eq 1).

Question 4

Build the model in the file coursework_model_template.slx by connecting the four pieces correctly:

Figure 3 shows a screenshot of the connected model. It is important to understand the functions of the blocks in order to connect the model correctly.

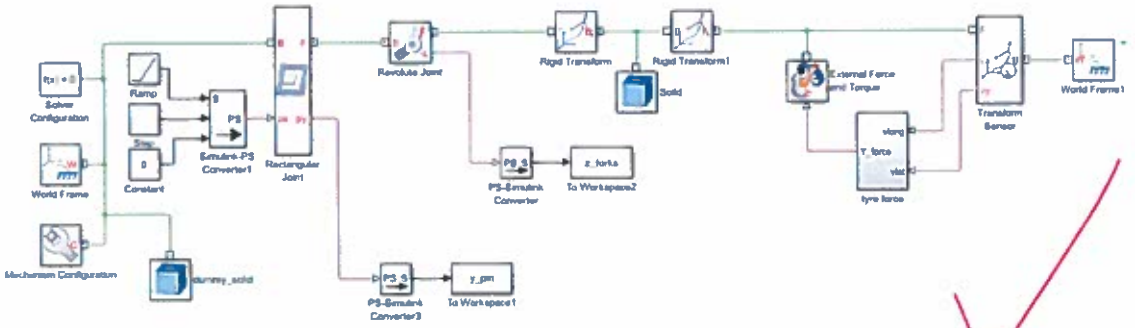


Figure 3: Screenshot of the completed SimMechanics model

Question 5

By examining the model, write the default values for e, f, m, I_z, and C:

The properties and settings of the model are explored to find the default values of the following parameters:

i) $e = 1\,m$
ii) $f = 0.6\,m$
iii) $m = 200\,kg$
iv) $I_z = 150\,kgm^2$
v) $C = 40000$

Question 6

Linearise the model and plot the eigenvalues of the system by varying the stiffness k of the king-pin mounting from 5000 to 100000 N/m:

The model is linearised using the linmod command. The eigenvalue plot is shown in Figure 4.

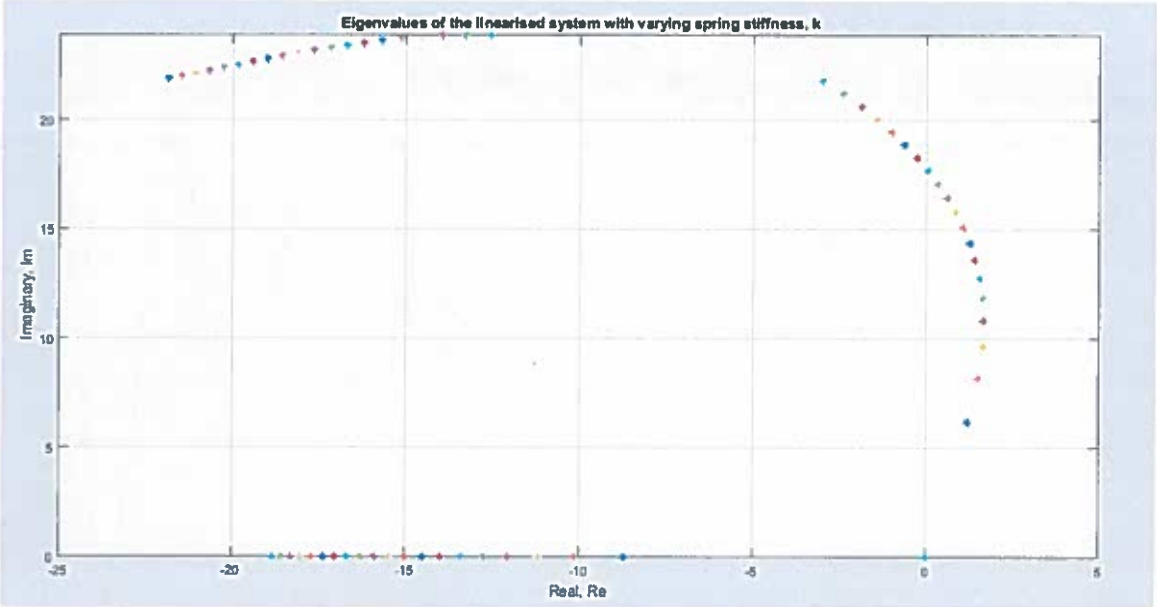


Figure 4: Eigenvalues of the linearised system with varying stiffness, k

As k increases, the eigenvalues become further from the positive side of the real axis. A system is unstable if it has poles in the positive side of the real axis. Hence, the stability of the system increases as k increases. From the plotting of the graph, we can observe that the eigenvalues are in the positive side of the real axis until the 13th point plotted, which corresponds to:

$k = 5000 * 13 = 65000 N/m.$

Therefore, we can conclude that the system is unstable when $k < 65000$.

Question 7

Verify two operating cases by running two simulations: $k = 30000$ and $k = 100000$. Plot the time response of the lateral displacement, y, and the z rotation of the forks for the two runs. Does the result of the simulation agree with the result of the linear eigenvalue analysis in terms of stability and frequency?

i) $k = 30000\ N/m$

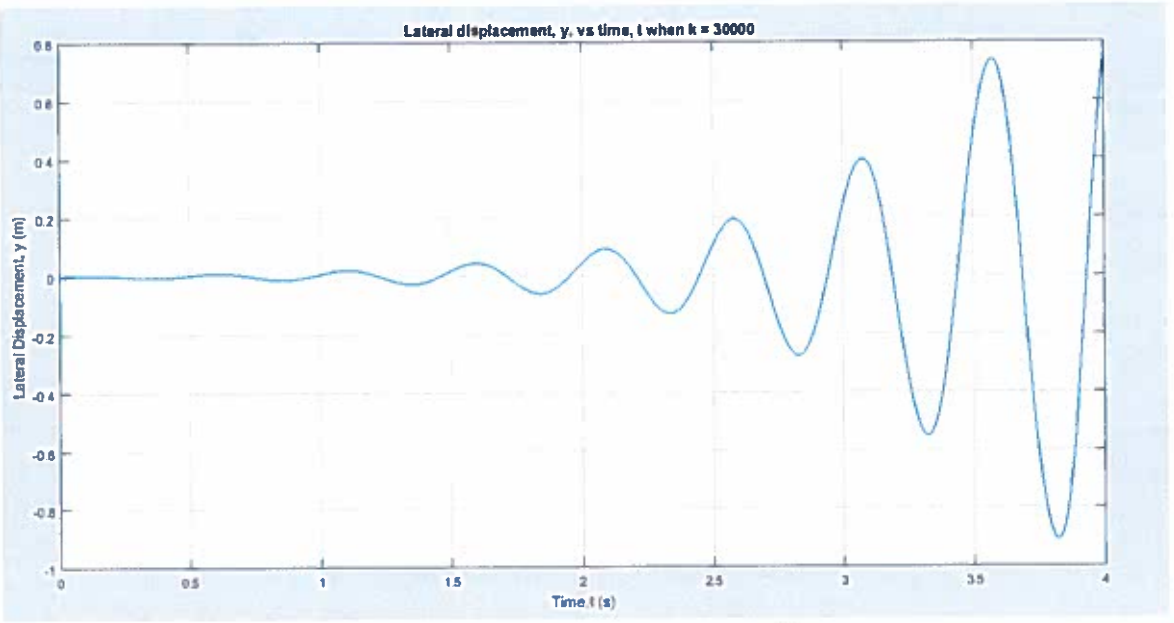


Figure 5: Lateral displacement, y, vs time, t when k = 30000

We can observe an increasing oscillation in the system, which indicates that the system is unstable. This gives the same result as the linear eigenvalue analysis, as the system is unstable when $k < 65000$. From the graph, we can calculate the frequency for y to be $2.01Hz$. In order to compare the frequencies between the two analysis, we must find the frequency from the eigenvalues. Figure 6 shows a table that contains all the eigenvalues, divided by 2π, in the system for $k = 30000$.

The frequency is the imaginary part of the eigenvalue divided by 2π. The real part of the eigenvalue gives the damping ratio of the component. The larger the value, the heavier the damping of that component. Therefore, the eigenvalue with the least value for the real part corresponds to the most dominant signal in the system. We can observe from the table in Figure 6 that the linearised system contains a dominant component with a frequency of $2.0328Hz$, which is very close to the frequency we obtained from the simulation. We are not able to observe the components which are heavily damped in the simulation results.

f_Z_A_30 | f_Z_A_100

9x1 complex double

	1
1	-3.1598 + 3.5885i
2	-3.1598 - 3.5885i
3	0.2462 + 2.0328i
4	0.2462 - 2.0328i
5	-2.1306 + 0.0000i
6	0.0000 + 0.0000i
7	0.0000 + 0.0000i
8	0.0000 + 0.0000i
9	0.0000 + 0.0000i
10	

Figure 6: Table f_Z_A_30 shows the eigenvalues of the system when k=30000, divided by 2π

The rotational displacement z is also showing instability in the system, as seen in figure 7. The frequency for z is calculated to be $2.02Hz$, which is very close to the frequency of the dominant signal in the system as well as the frequency for y.

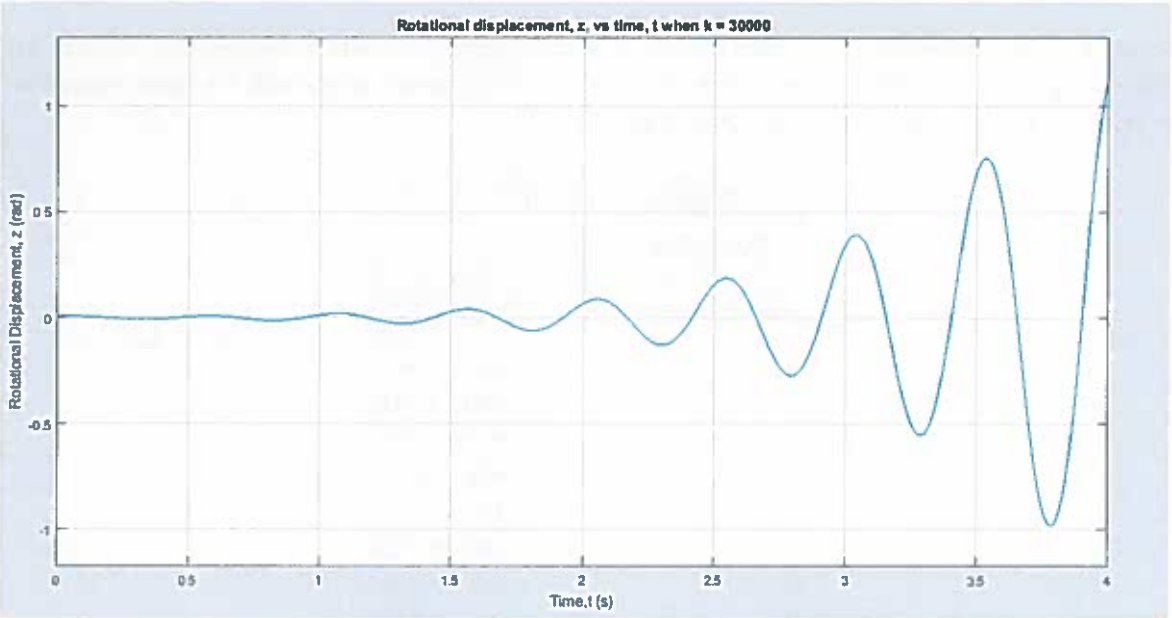


Figure 7: Rotational displacement, z, vs time, t when k = 30000

ii) $k = 100000\ N/m$

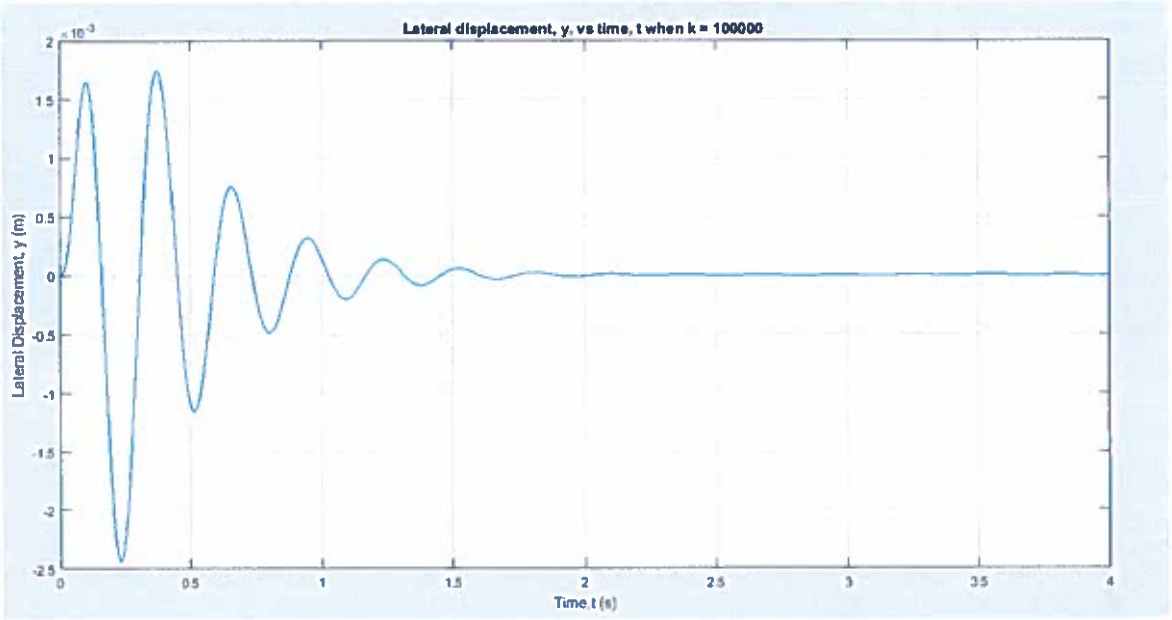


Figure 8: Lateral displacement,y, vs time,t, when k = 100000

Figure 8 shows a decreasing oscillation in the lateral displacement, y, with a frequency of $3.46Hz$. The decreasing oscillation observed shows that the system is stable, which agrees with the linear eigenvalue analysis, where the system becomes stable when $k > 65000$.

f_Z_A_30 | f_Z_A_100

9x1 complex double

	1
1	-2.9971 + 0.0000i
2	-2.0001 + 3.8041i
3	-2.0001 - 3.8041i
4	-0.4803 + 3.4737i
5	-0.4803 - 3.4737i
6	0.0000 + 0.0000i
7	0.0000 + 0.0000i
8	0.0000 + 0.0000i
9	0.0000 + 0.0000i
10	

Figure 9: Table f_Z_A_100 shows the eigenvalues of the system when k=100000, divided by 2π

From the table in Figure 9, we can see that the dominant signal of the system has a frequency of $3.4737Hz$, which is quite similar to the frequency calculated from the graph in Figure 8.

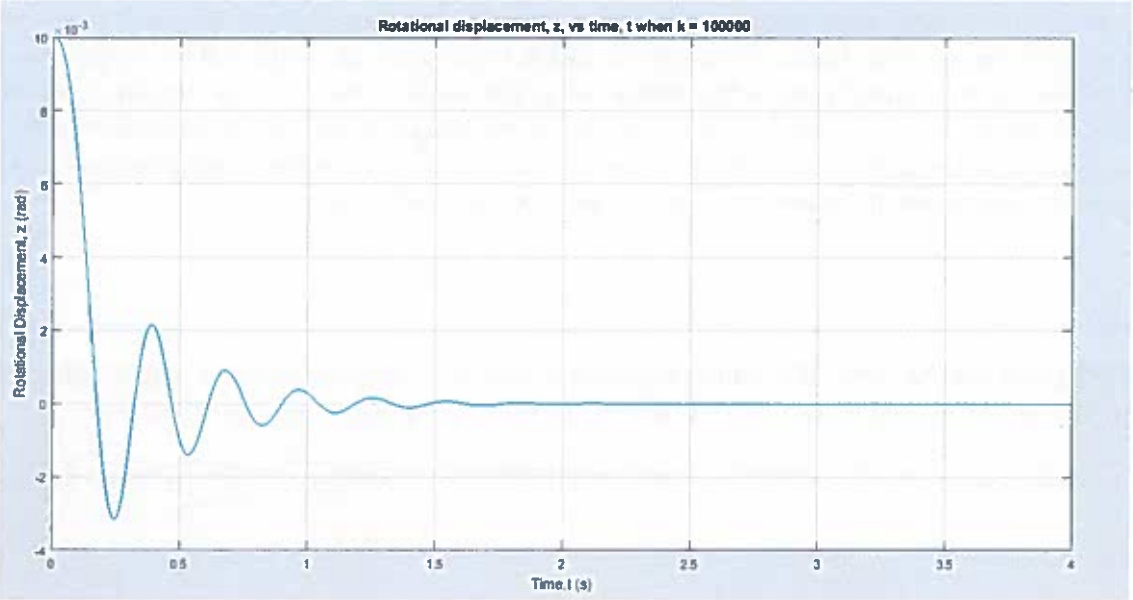


Figure 10: Rotational displacement, z, vs time, t, when k = 100000

As expected, the rotational displacement, z, shows a decreasing oscillation in Figure 10, making it a stable system. The frequency for z is calculated to be $3.48Hz$. As the first case, this value is nearly equal to the frequency of the dominant signal in the system and the frequency for y.

Question 8

Calculate and plot the eigenvalues of the linearised system by varying the forward speed from 1 to 50 m/s in steps of 1 m/s and fixing k to 100000 N/m:

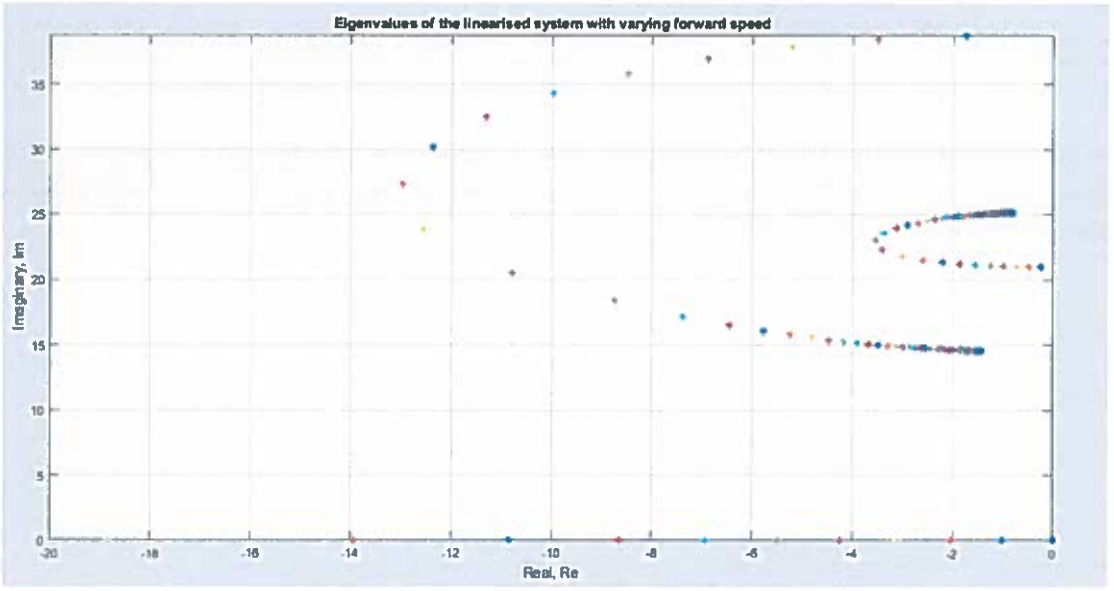


Figure 11: Eigenvalues of the linearised system with varying forward speed

The linmod command is again used to plot the eigenvalue of the linearised system. We can observe from the graph in figure 11 that there are no poles located at the positive side of the real axis, which means that there are no unstable poles in the system for all the varying forward speed. Therefore, we can conclude that the shimmy oscillations do not depend on the forward speed. As we can understand from the previous questions, the non-linearised system can be approximated by the linearised system, so we expect the same results if simulations were done with varying forward speed.

Question 9(a)

Calculate and plot the eigenvalues of the linearised system by varying the relaxation length, sigma_rel, from 0.02 to 1 m in steps of 0.02 m and fixing k to 100000 N/m and forward speed to 10 m/s:

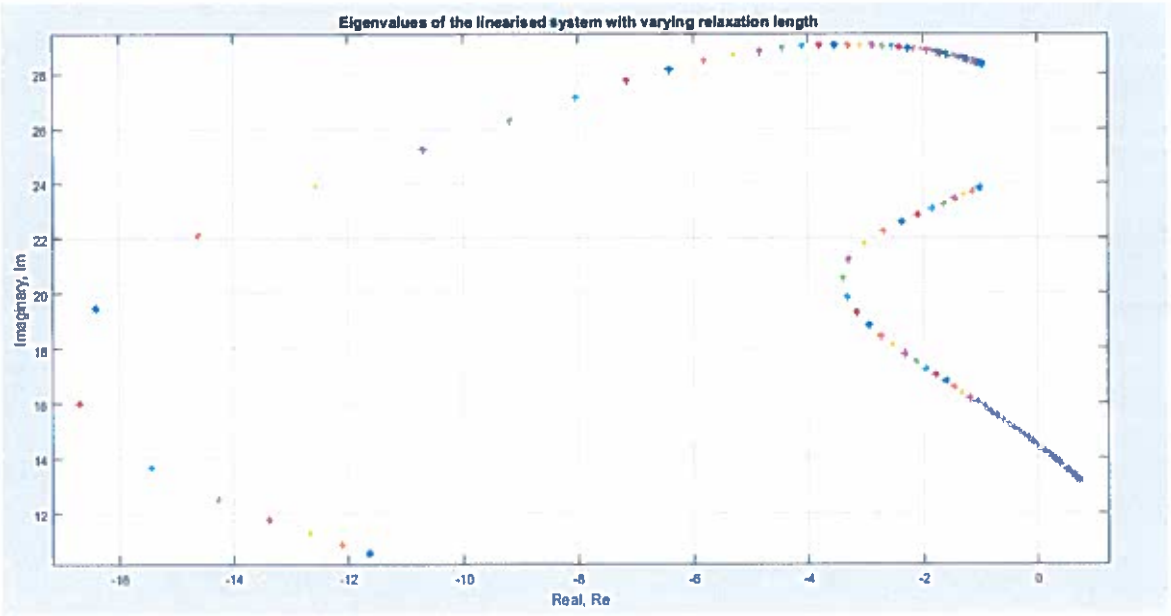


Figure 12: Eigenvalues of the linearised system with varying relaxation length, sigma_rel

From Figure 12, we can see that there are no poles on the positive side of the real axis up until a certain value of sigma_rel. To find the point at which the eigenvalue plot crosses to the positive side of the real axis, a zoomed-in version of the plot is shown in Figure 13.

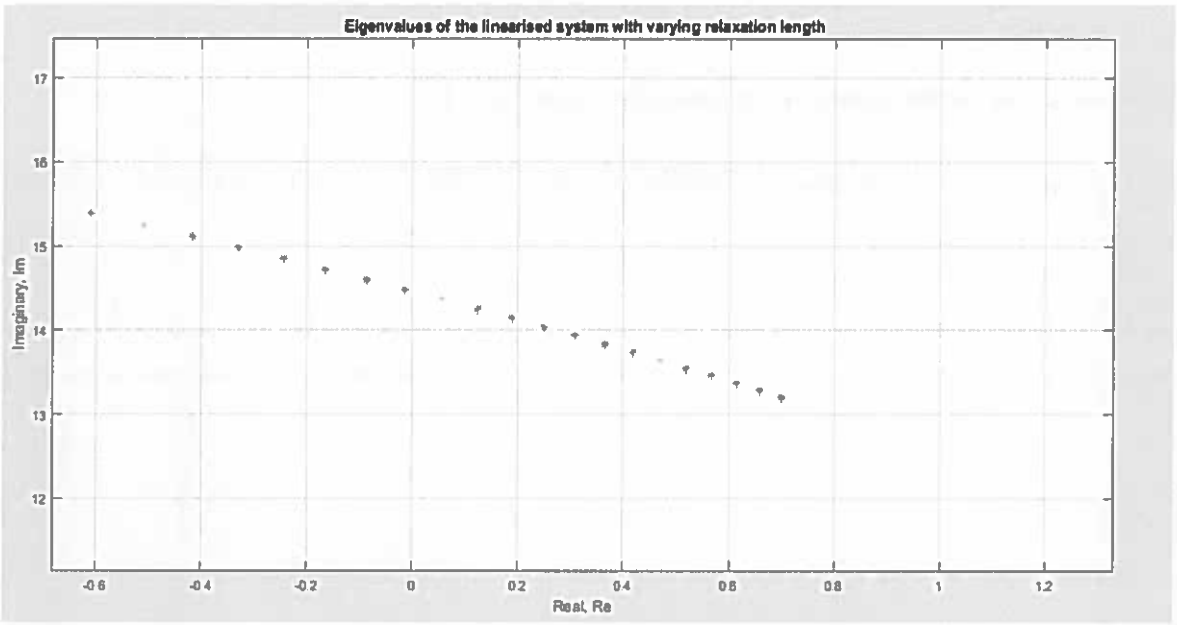


Figure 13: The eigenvalue plot zoomed-in at the imaginary axis

The point at which the eigenvalues are just about to cross to the positive side of the real axis is identified to be the 37th point plotted on the graph, which corresponds to the following value of sigma_rel:

$$sigma_rel = 0.02 * 37 = 0.74\, m.$$

Therefore, the system will remain in a stable state with changes in the relaxation length, sigma_rel, for sigma_rel $< 0.74\, m$.

Question 9(b)

Verify the stability of the system by simulation with sigma_rel $= 1\ m$:

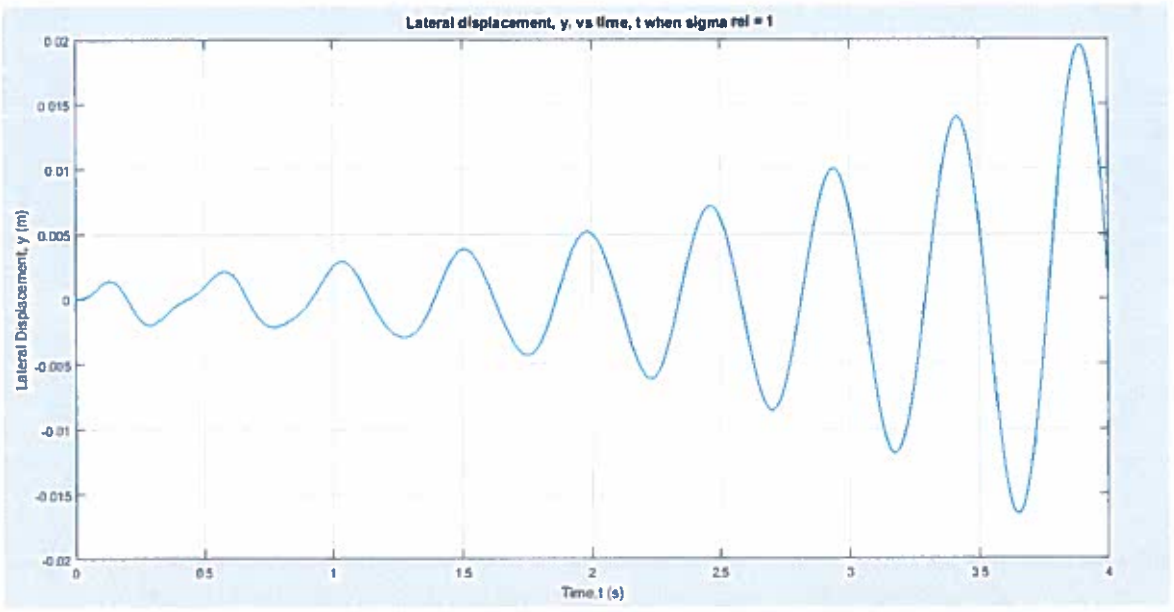


Figure 14: Lateral displacement, y, vs time, t, when sigma_rel = 1

Figure 14 shows the lateral displacement of the model as a result of the simulation. We can observe that the graph has an increasing oscillation, which indicates instability in the system, as suggested from the linear eigenvalue analysis in Question 9(a), where the system is unstable when sigma_rel $> 0.74\ m$.

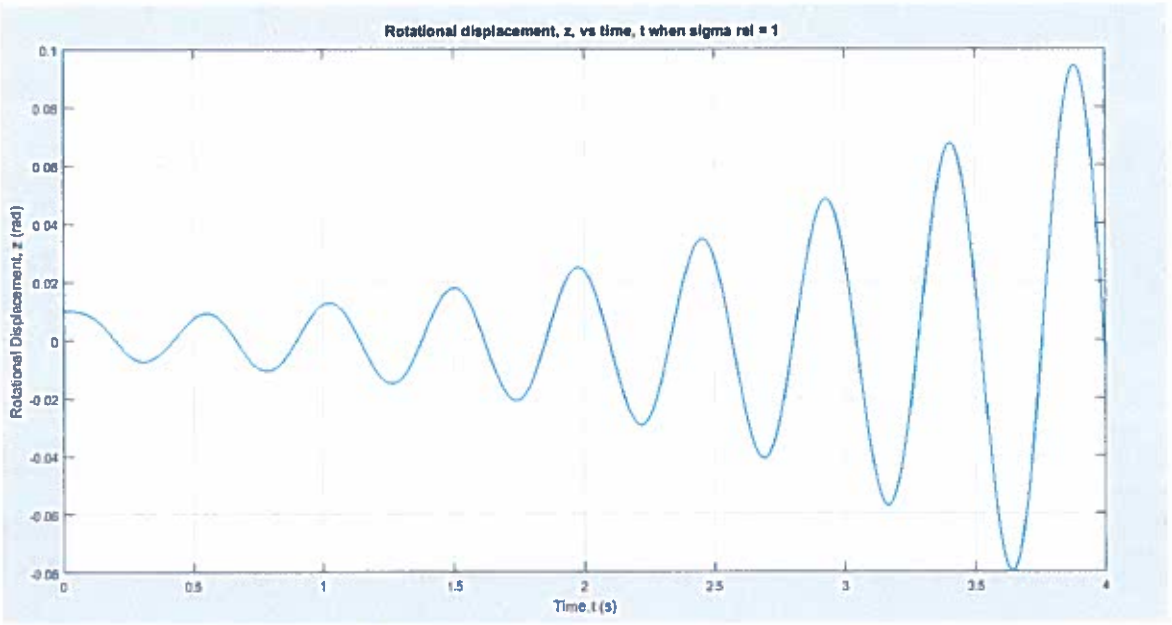


Figure 15: Rotational displacement, z, vs time, t, when sigma_rel = 1

As expected, the rotational displacement graph in Figure 15 also shows an increasing oscillation, therefore verifying the instability of the system as suggested from the eigenvalue plot in the linearised system.

Question 10

Question 10 consists of a few steps in order to achieve control and stabilise the wheel shimmy:

i) Calculate and plot the eigenvalues of the linearised system with the following default parameters:
   - $forward\ speed = 10\ m/s$
   - $sigma_rel = 0.2\ m$
   - $k = 60000\ N/m$

   Is the system unstable? Confirm the answer by running a simulation of the system.

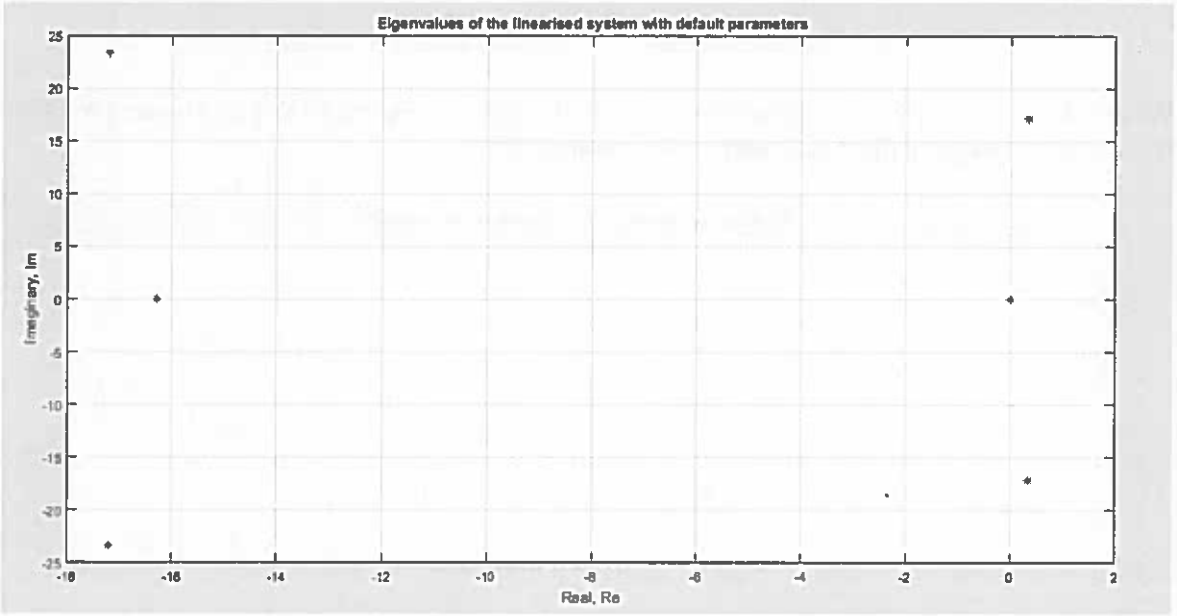


Figure 16: Eigenvalues of the linearised system with the default parameters

Figure 16 shows the eigenvalue plot of the linearised system with the default parameters given. As there are two poles which are located on the positive side of the real axis, we can conclude that the system is unstable with two unstable poles. The simulation of the model is done and the results are shown in Figures 17 and 18.

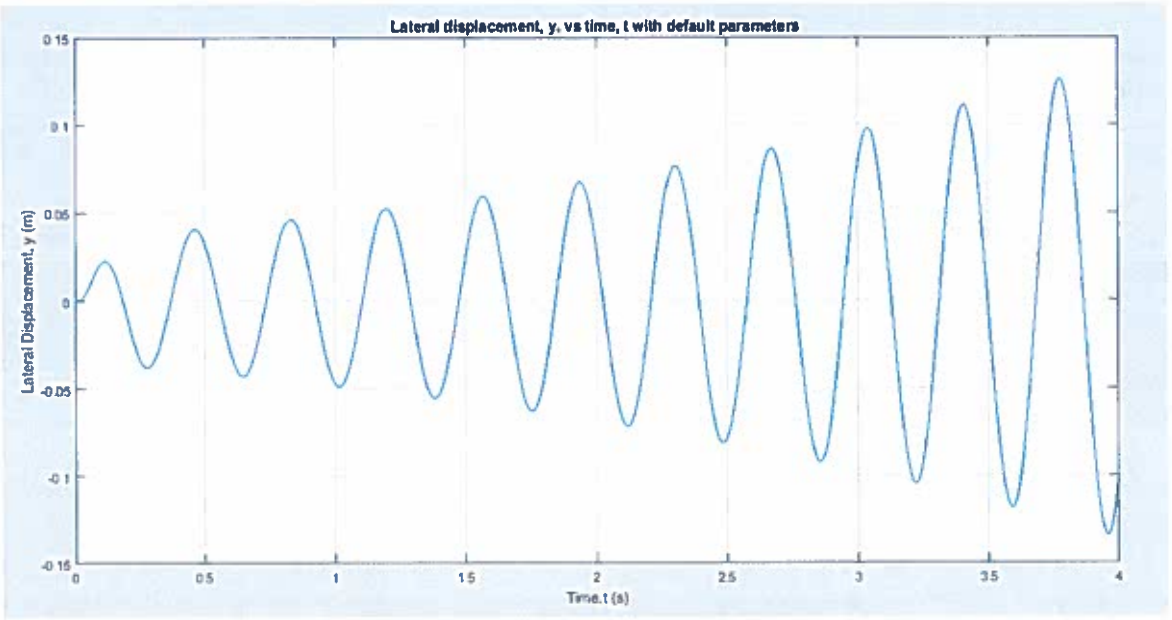


Figure 17: Lateral displacement, y, vs time, t, with the default parameters

We are able to observe increasing oscillations in both the graphs in Figures 17 and 18. This confirms that the system is unstable if the default parameters are applied to it.

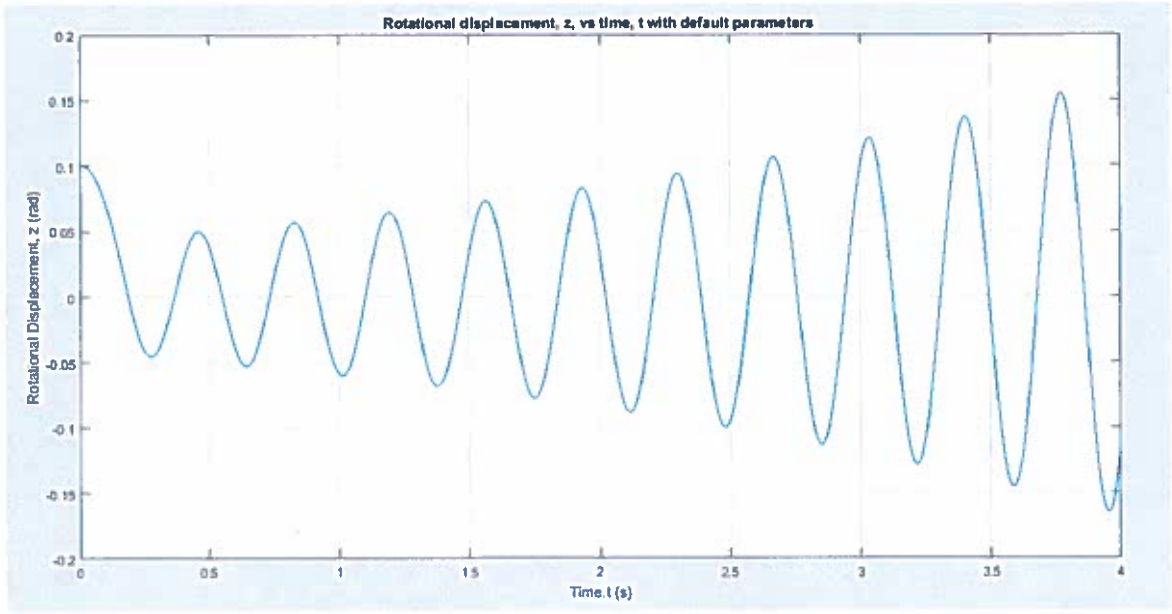


Figure 18: Rotational displacement, z, vs time, t, with the default parameters

ii) The system can be stabilised by applying the feedback control law as in Figure 19:

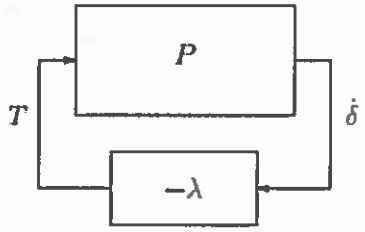


Figure 19: Feedback control law applied to a system P

Modify the model by adding an input corresponding to T and an output corresponding to $\dot{\delta}$

A screenshot of the new model is shown in Figure 20. The input port T is added as an actuation torque input to the revolute joint and the output port $\dot{\delta}$ is added as an output of the revolute joint measuring the steering velocity of the joint. The converters are required for the ports to convert Simulink inputs to a physical meaning for the system and vice versa.

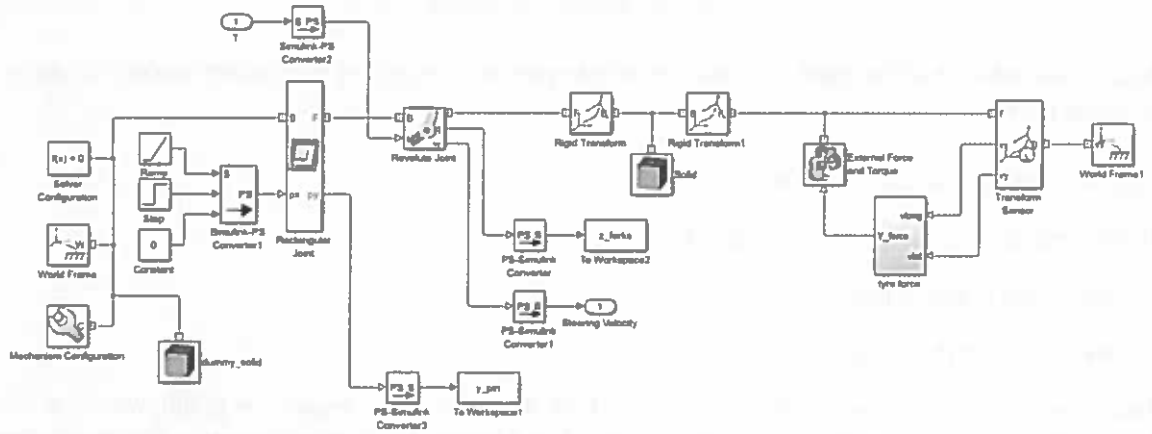


Figure 10: The new open-loop model

iii) Linearise the new open-loop model and plot the Nyquist diagram of the system. By using the Nyquist stability criterion, determine the value of λ that will stabilise the closed-loop system.

The following commands are used to complete the given task:

```
[A,B,C,D] = linmod('coursework_model_template');
sys = ss(A,B,C,D);
nyquistplot(sys);
```

Figure 21 shows the Nyquist diagram obtained for the open-loop system.

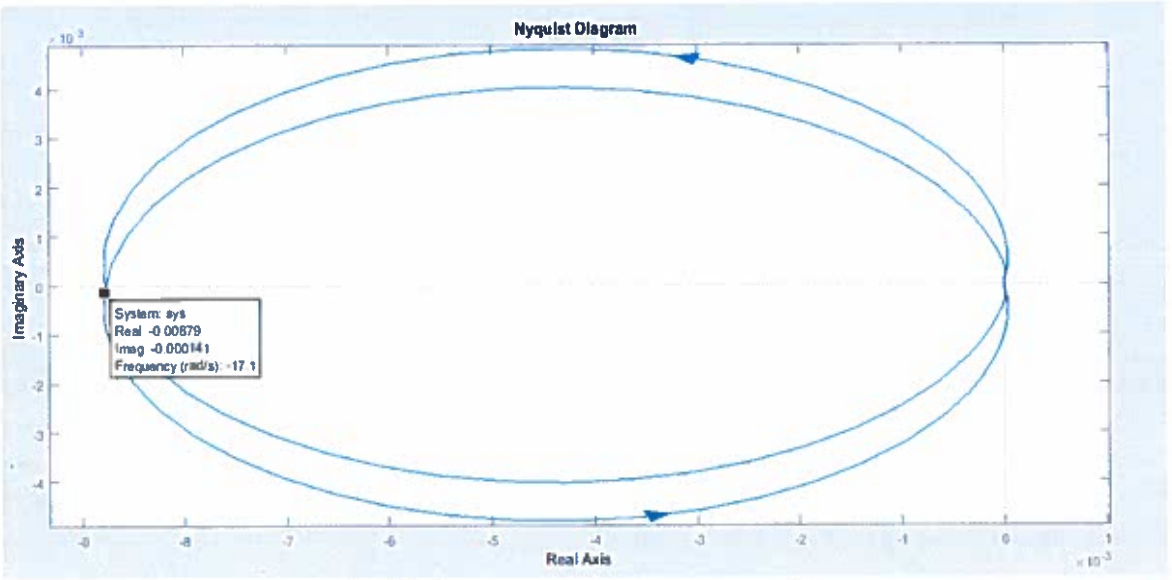


Figure 11: Nyquist diagram of the open-loop system

In order to understand the stability properties of the system, we must use the Nyquist stability criterion to inspect the plot.

Nyquist stability criterion: N = Z - P

N: No. of clockwise encirclements about -1

Z: Closed-loop unstable poles

P: Open-loop unstable poles

From the graph, there is no encirclement about -1, so N = 0. From the eigenvalue graph, we know that there are two unstable poles in the system, so P = 2. Therefore, Z = 2. This shows that the closed-loop system is unstable. We can scale the Nyquist diagram up and down by adding a gain, K, in the closed-loop system. This modifies the criterion, therefore N is now the number of clockwise encirclements about -1/K.

From the criterion, the system is stable if and only if N = -P. In this system, N must be equal to -2. To achieve a marginal stability, N must marginally encircle the point -1/K twice in the anticlockwise direction (anticlockwise gives a negative value to N). To choose the value K, we do the calculation as follows:

$$-\frac{1}{K} = 0.00879$$

$$K = -114$$

Therefore, $\lambda = 114$.

iv) Implement the control law in the model and initialise the z rotational displacement at $0.1\ rad$. Run a simulation to confirm that the chosen λ produces a marginal stability.

Figure 22 shows a screenshot of the closed-loop model. The T and $\dot{\delta}$ ports are removed in order to connect the input and output ports in a loop, with a gain of $-\lambda$, where λ is 114.

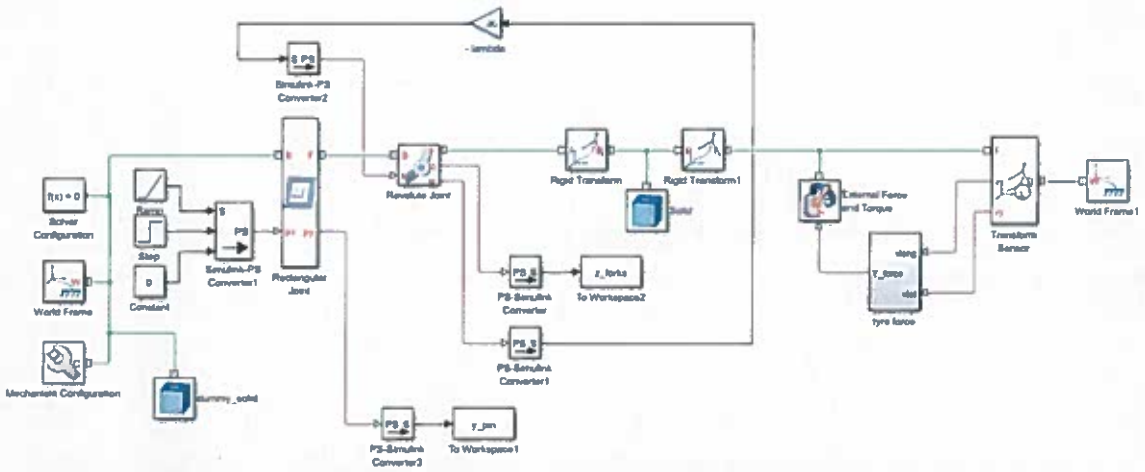


Figure 12: A screenshot of the closed-loop model

The results from the simulation of the closed-loop model are shown in Figures 23 and 24.

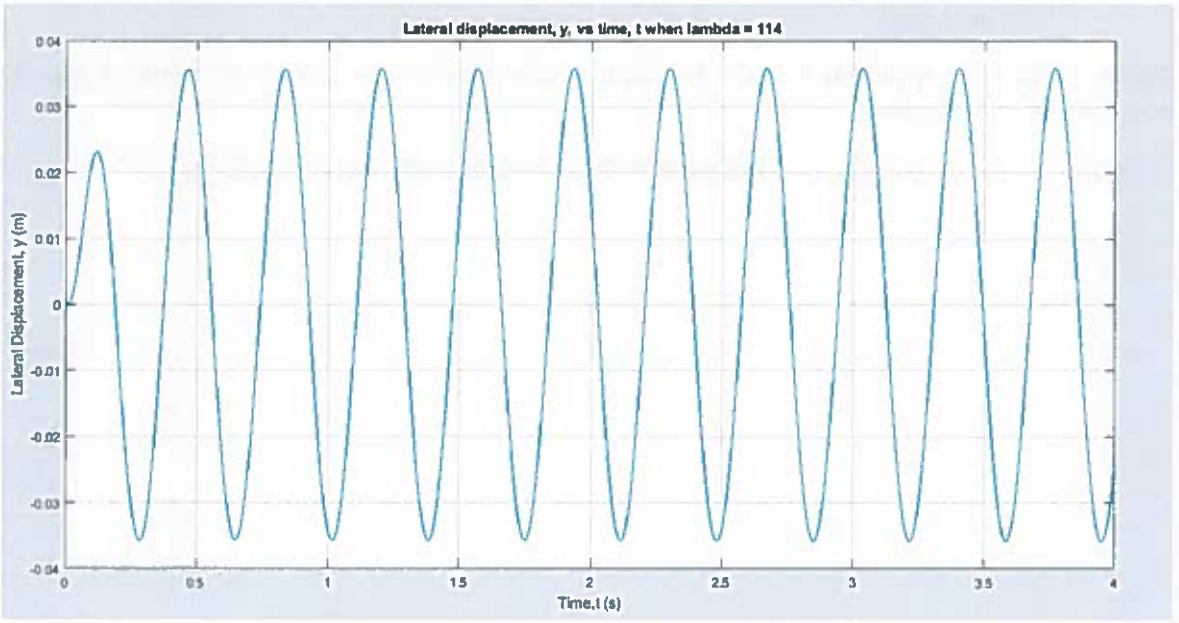


Figure 13: Lateral displacement, y, vs time, t, when lambda = 114

It can be seen from both displacement graphs that the system has a sustained oscillation, therefore confirming that the closed-loop system is marginally stable when $\lambda = 114$.

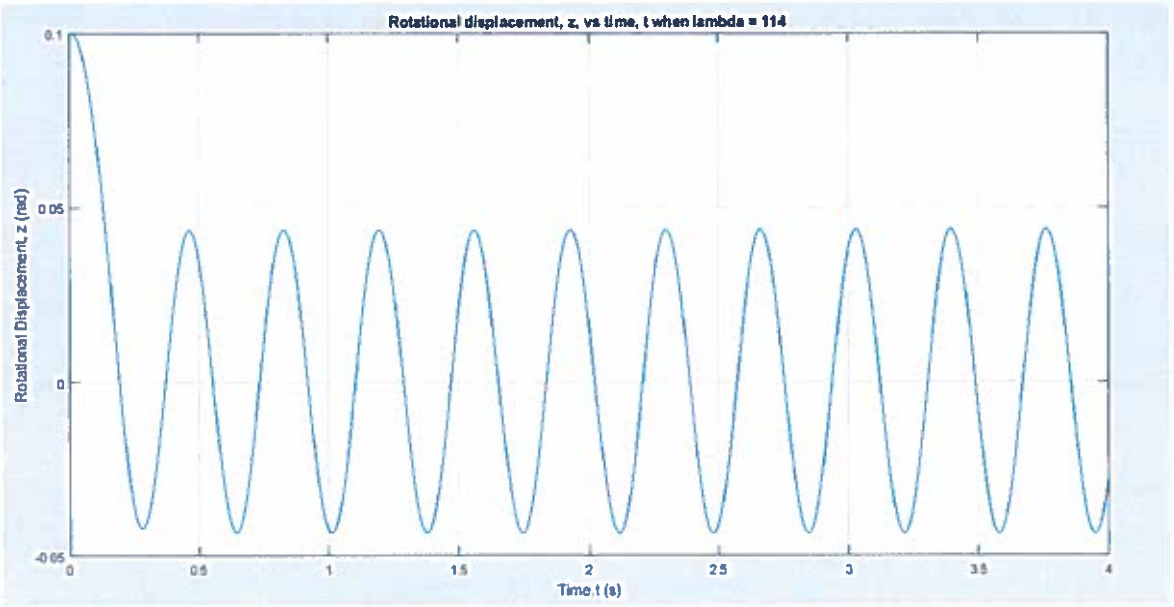


Figure 14: Rotational displacement, z, vs time, t, when lambda = 114

v) Run a simulation of the closed-loop model with a larger value of λ to confirm that the system will be stable.

The simulation is run again with $\lambda = 600$. Figures 25 and 26 show the results of the simulation in regards to the model's displacements.

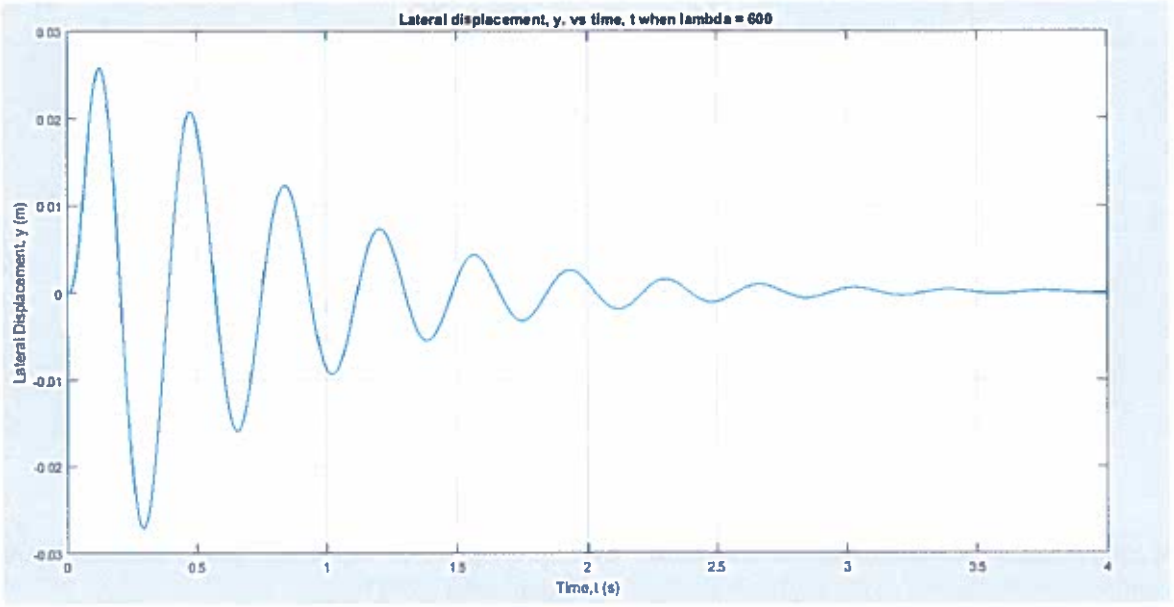


Figure 15: Lateral displacement, y, vs time, t, when lambda = 600

We can observe a decreasing oscillation in both displacement graphs, and this verifies that the closed-loop system is stable if $\lambda > 114$.

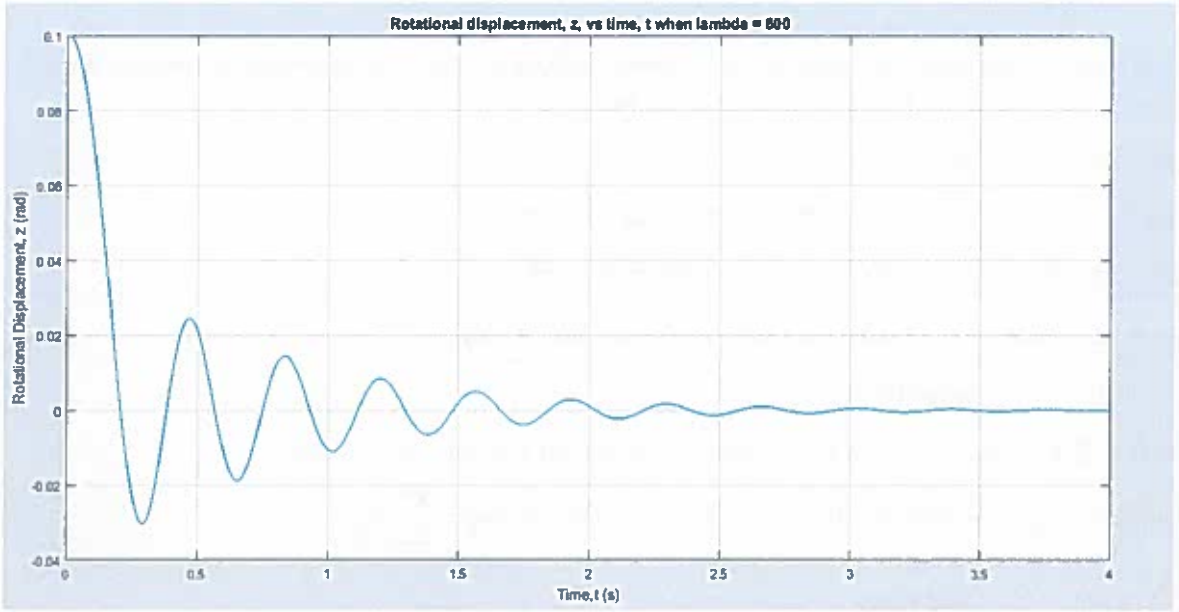


Figure 16: Rotational displacement, z, vs time, t, when lambda = 600

Conclusion

From all the tasks completed, we can conclude that a physical system can be stabilised by applying the feedback control law and analysing its Nyquist diagram. This control method is very useful in mechanical bodies such as vehicles and machineries to ensure they remain stable when in use. Having instability and oscillations in a physical system, such as a vehicle, will be very dangerous and may cause accidents and fatalities. Therefore, it is important for all mechanical bodies to be designed as stable as possible.

We also learnt that linearisation can be used to approximate this non-linearised system and investigate the outcome of the system almost accurately. This saves us a lot of time in finding the best solutions to stabilise the system. It is also helps us in understanding how varying parameter values may affect the stability of the model.

Overall, the coursework helps us to understand how we are able to use the feedback control law in order to stabilise a physical system in real-life situations. This basic idea can be applied to further complicated models and help us design more stable systems in the industry. This knowledge can also be used to recognise the cause of accidents and failures that may occur in mechanical bodies.

Appendix A

This appendix contains the derivation for question 1 using the Newtonian approach for comparison with the Lagrangian approach, which produces equal results:

$$\ddot{\boldsymbol{R}} = \ddot{x}\boldsymbol{i} + \ddot{y}\boldsymbol{j} - f\ddot{\delta}\boldsymbol{e_\theta} + f\dot{\delta}^2\boldsymbol{e_r}$$

$$\boldsymbol{e_\theta} = -\sin\delta\,\boldsymbol{i} + \cos\delta\,\boldsymbol{j} \qquad \text{and} \qquad \boldsymbol{e_r} = \cos\delta\,\boldsymbol{i} + \sin\delta\,\boldsymbol{j}$$

$$\ddot{\boldsymbol{R}} = \ddot{x}\boldsymbol{i} + \ddot{y}\boldsymbol{j} - f\ddot{\delta}(-\sin\delta\,\boldsymbol{i} + \cos\delta\,\boldsymbol{j}) + f\dot{\delta}^2(\cos\delta\,\boldsymbol{i} + \sin\delta\,\boldsymbol{j})$$

$$= \left(\ddot{x} + f\ddot{\delta}\sin\delta + f\dot{\delta}^2\cos\delta\right)\boldsymbol{i} + (\ddot{y} - f\ddot{\delta}\cos\delta + f\dot{\delta}^2\sin\delta)\boldsymbol{j}$$

Using the following equation:

$$m\ddot{\boldsymbol{R}} = \textstyle\sum_i \boldsymbol{F}_i^{(e)} \equiv \boldsymbol{F}^{(e)} \textit{ where } \boldsymbol{F}^{(e)} \textit{ is the total external force on the system}$$

$$m\left[\left(\ddot{x} + f\ddot{\delta}\sin\delta + f\dot{\delta}^2\cos\delta\right)\boldsymbol{i} + (\ddot{y} - f\ddot{\delta}\cos\delta + f\dot{\delta}^2\sin\delta)\boldsymbol{j}\right] = \sum_i \boldsymbol{F}_i^{(e)}$$

$$\sum_i \boldsymbol{F}_i^{(e)} = F_x\boldsymbol{i} - ky\boldsymbol{j} + Y\boldsymbol{e_\theta}$$

$$= (F_x - Y\sin\delta)\boldsymbol{i} + (-ky + Y\cos\delta)\boldsymbol{j}$$

Therefore, the first two equations of motion are as follows:

$$m\left(\ddot{x} + f\ddot{\delta}\sin\delta + f\dot{\delta}^2\cos\delta\right) = F_x - Y\sin\delta \qquad (1)$$

$$m\left(\ddot{y} - f\ddot{\delta}\cos\delta + f\dot{\delta}^2\sin\delta\right) = -ky + Y\cos\delta \qquad (2)$$

$$\boldsymbol{N} = \frac{d\boldsymbol{H}}{dt}$$

$$I_z\ddot{\delta}\boldsymbol{k} = f\boldsymbol{e_r} \times (F_x\boldsymbol{i} - ky\boldsymbol{j}) - (e - f)\boldsymbol{e_r} \times Y\boldsymbol{e_\theta}$$

$$I_z\ddot{\delta}\boldsymbol{k} = \left[(-f\sin\delta\left(m\left(\ddot{x} + f\ddot{\delta}\sin\delta + f\dot{\delta}^2\cos\delta\right) + Y\sin\delta\right) + f\cos\delta\left(m\left(\ddot{y} - f\ddot{\delta}\cos\delta + f\dot{\delta}^2\sin\delta\right) - Y\cos\delta\right) - (e - f)Y\right]\boldsymbol{k}$$

$$I_z\ddot{\delta} = -mf\ddot{x}\sin\delta + mf\ddot{y}\cos\delta - mf^2\ddot{\delta} - eY$$

Therefore, the third equation of motion is:

$$(I_z + mf^2)\ddot{\delta} + m(f\ddot{x}\sin\delta - f\ddot{y}\cos\delta) = -eY \qquad (3)$$



Appendix B

This appendix contains all the Matlab commands used to complete questions 6-10 of the coursework:

Question 6

```
z_initial = 0;
speed = 10;
sigma_rel = 0.2;

for i = 1:20
   k = 5000*i;
   [A,B,C,D] = linmod('coursework_model_template');
   Z_A = eig(A);
   real_Z_A = real(Z_A);
   imag_Z_A = imag(Z_A);
   plot(real_Z_A,imag_Z_A,'*');
   ylim([0,+inf]);
   hold on;
   xlabel('Real, Re');
   ylabel('Imaginary, Im');
   title('Eigenvalues of the linearised system with varying spring stiffness, k');
   if k == 30000;
         f_Z_A_30 = Z_A/(2*pi);
   end
   if k == 100000;
         f_Z_A_100 = Z_A/(2*pi);
   end
end
```

Question 7(a)

```
z_initial = 0.01;
speed = 10;
sigma_rel = 0.2;
k = 30000;

sim('coursework_model_template');
figure;
plot(y_pin.time,y_pin.signals.values);
xlabel('Time,t (s)');
ylabel('Lateral Displacement, y (m)');
title('Lateral displacement, y, vs time, t when k = 30000');

figure;
```

```
plot(z_forks.time,z_forks.signals.values);
xlabel('Time,t (s)');
ylabel('Rotational Displacement, z (rad)');
title('Rotational displacement, z, vs time, t when k = 30000');
```

Question 7(b)

```
z_initial = 0.01;
speed = 10;
sigma_rel = 0.2;
k = 100000;

sim('coursework_model_template');
figure;
plot(y_pin.time,y_pin.signals.values);
xlabel('Time,t (s)');
ylabel('Lateral Displacement, y (m)');
title('Lateral displacement, y, vs time, t when k = 100000');

figure;
plot(z_forks.time,z_forks.signals.values);
xlabel('Time,t (s)');
ylabel('Rotational Displacement, z (rad)');
title('Rotational displacement, z, vs time, t when k = 100000');
```

Question 8

```
z_initial = 0;
sigma_rel = 0.2;
k = 100000;

for i = 1:50
   speed = i;
   [A,B,C,D] = linmod('coursework_model_template');
   Z_A = eig(A);
   real_Z_A = real(Z_A);
   imag_Z_A = imag(Z_A);
   plot(real_Z_A,imag_Z_A,'*');
   axis([-20,+inf,0,+inf]);
   hold on;
   xlabel('Real, Re');
   ylabel('Imaginary, Im');
   title('Eigenvalues of the linearised system with varying forward speed');
end
```

Question 9(a)

```
z_initial = 0;
speed = 10;
k = 100000;

for i = 1:50
   sigma_rel = 0.02*i;
   [A,B,C,D] = linmod('coursework_model_template');
   Z_A = eig(A);
   real_Z_A = real(Z_A);
   imag_Z_A = imag(Z_A);
   plot(real_Z_A,imag_Z_A,'*');
   ylim([0,+inf]);
   hold on;
   xlabel('Real, Re');
   ylabel('Imaginary, Im');
   title('Eigenvalues of the linearised system with varying relaxation length');
end
```

Question 9(b)

```
z_initial = 0.01;
speed = 10;
sigma_rel = 1;
k = 100000;

sim('coursework_model_template');
figure;
plot(y_pin.time,y_pin.signals.values);
xlabel('Time,t (s)');
ylabel('Lateral Displacement, y (m)');
title('Lateral displacement, y, vs time, t when sigma_rel = 1');
figure;
plot(z_forks.time,z_forks.signals.values);
xlabel('Time,t (s)');
ylabel('Rotational Displacement, z (rad)');
title('Rotational displacement, z, vs time, t when sigma_rel = 1');
```

Question 10

```
z_initial = 0;
speed = 10;
sigma_rel = 0.2;
k = 60000;
```

```
[A,B,C,D] = linmod('coursework_model_template');
Z_A = eig(A);
real_Z_A = real(Z_A);
imag_Z_A = imag(Z_A);
figure;
plot(real_Z_A,imag_Z_A,'*');
xlabel('Real, Re');
ylabel('Imaginary, Im');
title('Eigenvalues of the linearised system with default parameters');

z_initial = 0.1;
sim('coursework_model_template');
figure;
plot(y_pin.time,y_pin.signals.values);
xlabel('Time,t (s)');
ylabel('Lateral Displacement, y (m)');
title('Lateral displacement, y, vs time, t with default parameters');
figure;
plot(z_forks.time,z_forks.signals.values);
xlabel('Time,t (s)');
ylabel('Rotational Displacement, z (rad)');
title('Rotational displacement, z, vs time, t with default parameters');

[A,B,C,D] = linmod('coursework_model_template');
sys = ss(A,B,C,D);
nyquistplot(sys);

z_initial = 0.1;
lambda = 114;
sim('coursework_model_template');
figure;
plot(y_pin.time,y_pin.signals.values);
xlabel('Time,t (s)');
ylabel('Lateral Displacement, y (m)');
title('Lateral displacement, y, vs time, t when lambda = 114');
figure;
plot(z_forks.time,z_forks.signals.values);
xlabel('Time,t (s)');
ylabel('Rotational Displacement, z (rad)');
title('Rotational displacement, z, vs time, t when lambda = 114');

z_initial = 0.1;
lambda = 600;
sim('coursework_model_template');
```

```
figure;
plot(y_pin.time,y_pin.signals.values);
xlabel('Time,t (s)');
ylabel('Lateral Displacement, y (m)');
title('Lateral displacement, y, vs time, t when lambda = 600');
figure;
plot(z_forks.time,z_forks.signals.values);
xlabel('Time,t (s)');
ylabel('Rotational Displacement, z (rad)');
title('Rotational displacement, z, vs time, t when lambda = 600');
```